

Twin Pregnancy 02. Maternal Physiology and Global Risk

Companion reading handout for simultaneous listening.

Audio course on twin pregnancy. Episode two. Maternal physiology and global risk.

Source base

This episode is based on Gabbe chapters on multiple gestations, maternal physiology, hematologic disease, gastrointestinal disease, preeclampsia, and the Israeli position papers and protocols on twin follow-up, nutrition, hypertension, venous thromboembolism, and postpartum hemorrhage.

Opening frame

Let us begin with the deepest idea.

Twin pregnancy is not just a singleton pregnancy plus one fetus. It is an amplification of normal pregnancy physiology. The same adaptations that are usually tolerable in singleton pregnancy are pushed closer to their limits.

That is why twins increase the risk of almost every obstetric complication except the complications that depend on prolonged gestation or excessive fetal size, such as postterm pregnancy and macrosomia.

Start with the maternal cardiovascular system

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Pregnancy normally increases plasma volume, heart rate, stroke volume, and cardiac output. Gabbe notes that cardiac output rises substantially in normal pregnancy, and twin gestation adds another increment above singleton pregnancy. Clinically, this matters because the mother is living with more volume, more venous capacitance, more edema tendency, and less reserve if disease appears.

Now connect that to pulmonary edema

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Twin pregnancy increases total body water and lowers colloid oncotic pressure. Lower oncotic pressure means fluid more easily leaves the intravascular

compartment. Add preeclampsia, tocolysis, excessive intravenous fluids, corticosteroids, anemia, or cardiac disease, and the patient can cross the threshold into pulmonary edema.

So when a twin patient with preterm contractions is receiving fluids and medication, the question is not only, can we stop contractions. It is also, how much cardiopulmonary reserve are we consuming.

Next, the endocrine and metabolic load.

More placental mass means more placental hormone production. Human placental lactogen and related signals increase insulin resistance. That is one reason twin pregnancies have a higher risk of gestational diabetes. The fetus is not simply demanding calories. The placenta is actively changing maternal metabolism.

Multiple gestation is also associated with hyperemesis gravidarum. The mechanism is not one molecule only, but higher placental hormone burden and early pregnancy endocrine load are clinically plausible contributors. When severe vomiting is evaluated, multiple gestation and molar pregnancy are classic diagnoses to exclude.

Now anemia and nutrition

Now anemia and nutrition.

Twin pregnancy expands plasma volume more, increases iron demand, and increases folate demand. The Israeli twin follow-up paper recommends folic acid and iron supplementation throughout pregnancy. The nutrition position paper gives twin-specific weight gain ranges by pre-pregnancy body mass index. For a normal body mass index, the Institute of Medicine range is 17 to 25 kilograms. For overweight, 14 to 23 kilograms. For obesity, 11 to 19 kilograms.

But the teaching point is not the table. The teaching point is growth reserve. In twins, inadequate maternal nutritional gain is linked to fetal growth problems and preterm birth, while excessive gain can worsen diabetes, hypertension, and operative risk. Nutrition is not cosmetic counseling. It is placental and neonatal risk management.

Now hypertensive disease

Now hypertensive disease.

Twin pregnancy is a risk factor for preeclampsia. The Israeli hypertension and twin follow-up documents both treat multiple pregnancy as a meaningful risk factor. Low-dose aspirin should be considered from the end of the first trimester until delivery when additional risk factors exist, such as nulliparity, age 40 or above, body mass index 35 or above, and no contraindication.

Why is preeclampsia more common? Think placental mass, placental ischemic burden, angiogenic imbalance, and maternal endothelial susceptibility. A twin placenta, or two placentas, increases the total placental interface. More placental tissue can mean more placental signal. If placentation is abnormal, that signal becomes antiangiogenic and inflammatory.

Now hemorrhage

Now hemorrhage.

Twin pregnancy predisposes to postpartum hemorrhage largely through uterine overdistension. The myometrium that has been stretched by two fetuses and more fluid may contract less efficiently after delivery. Add operative delivery, induction or augmentation, chorioamnionitis, prolonged labor, retained placenta, or anemia, and hemorrhage risk rises further.

This is why twin delivery preparation includes intravenous access, blood availability, active management of the third stage, and readiness for uterotonics. These are not checklist rituals. They are responses to uterine mechanics.

Now venous thromboembolism, or VTE

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Pregnancy is already hypercoagulable. Twin pregnancy adds immobility risk, more frequent hospitalization, preeclampsia, cesarean delivery, postpartum hemorrhage, and sometimes fertility-treatment background. The local VTE protocol lists multiple gestation as a risk factor. The practical point is that twin pregnancy should raise your threshold for formal VTE risk assessment, especially antepartum admissions and postpartum.

Let us connect the maternal risks into one model.

Twin pregnancy increases volume load, metabolic load, placental signaling, uterine stretch, preterm-birth risk, and procedural exposure. Those loads explain most maternal complications: anemia, hyperemesis, diabetes, preeclampsia, pulmonary edema, thromboembolism, operative delivery, and hemorrhage.

The pathophysiology also explains why twin clinics and structured surveillance are useful. They do not merely detect fetal problems. They repeatedly assess maternal blood pressure, symptoms, growth, nutrition, cervix, and delivery planning before the physiology destabilizes.

Pause and summarize.

Twins amplify normal pregnancy adaptation. More blood volume stresses the heart. More placental mass changes metabolism and vascular signaling. More uterine

stretch increases preterm birth and hemorrhage risk. More clinical complexity increases intervention and operative exposure.

Now final synthesis

Now final synthesis.

Key point

First, twin pregnancy is high risk because normal maternal physiology is pushed harder.

Second, pulmonary edema risk is understandable when high cardiac output, expanded volume, lower oncotic pressure, preeclampsia, tocolysis, and fluids meet in the same patient.

Third, preeclampsia, diabetes, anemia, hyperemesis, VTE, and postpartum hemorrhage are not random associations. They follow from placental mass, metabolic demand, vascular signaling, and uterine overdistension.

Fourth, maternal preparation for twin delivery begins long before the delivery room. It begins with nutrition, blood pressure risk assessment, anemia prevention, VTE awareness, and a low threshold to recognize decompensation.

End of episode two.